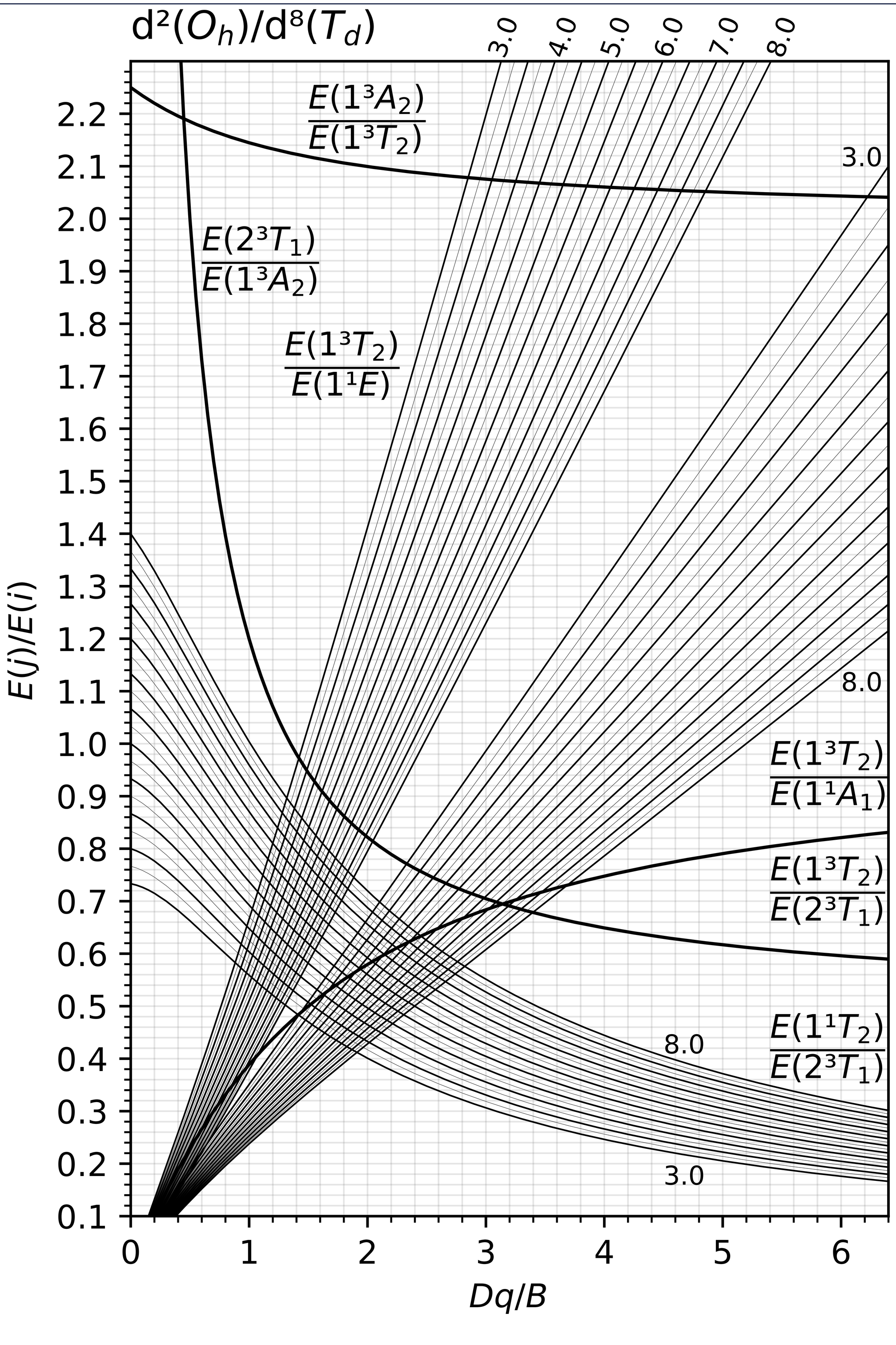
Auxiliary Tanabe–Sugano Diagram



Electrostatic matrix elements $\alpha\hbar c\langle\Gamma; t_2^n e^m|\,\Sigma_{i<j}\, r_{ij}^{-1}|\Gamma; t_2^{n'} e^{m'}\rangle$

$^1A_1(^1\mathrm{G}, ^1\mathrm{S})$:

t_2^2	e^2
$10B+5C$	$\sqrt{6}(2B+C)$
	$8B+4C$

$^1E(^1\mathrm{D}, ^1\mathrm{G})$:

t_2^2	e^2
$B+2C$	$-2\sqrt{3}B$
	$2C$

$^1T_2(^1\mathrm{D}, ^1\mathrm{G})$:

t_2^2	$t_2{}^1e^1$
$B+2C$	$2\sqrt{3}C$
	$2C$

$^3T_1(^3\mathrm{F}, ^3\mathrm{P})$:

t_2^2	$t_2^1e^1$
$-5B$	$6B$
	$4B$

$^1T_1(^1\mathrm{G})$: $t_2^1e^1,\ 4B+2C$
$^3T_2(^3\mathrm{F})$: $t_2^1e^1,\ -8B$
$^3A_2(^3\mathrm{F})$: $e^2,\ -8B$